



SKG14 IC8 Pacing and Emerging Technologies for ICM Engineers, L2

5.0 IIoT Use Cases

July 2025

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Learning Outcome

At the end of the session, you will be able to:

- Understand the importance of integrating automated systems to streamline workflows and transactions for industrial and business processes.
- Explore technologies enabling efficient real-time monitoring and control of devices and systems, improving operational efficiency.
- Gain insights into how AI and analytics can enhance decision-making, operational performance, and predictive capabilities.
- Learn about practical implementations of IoT and analytics in promoting sustainable practices across industries.

05. IIoT Use Cases

- Transactional and Workflow Based Application
- Monitoring and Remote Control
- Advanced Analytics/AI Application
- Sustainability Use Cases

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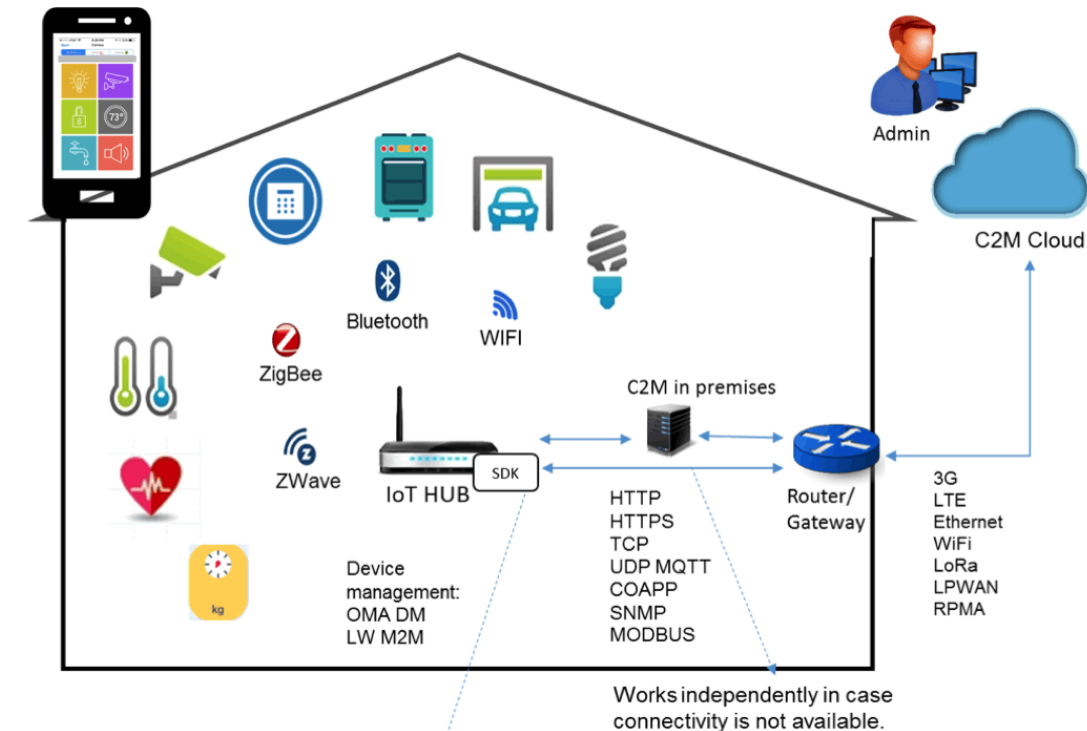
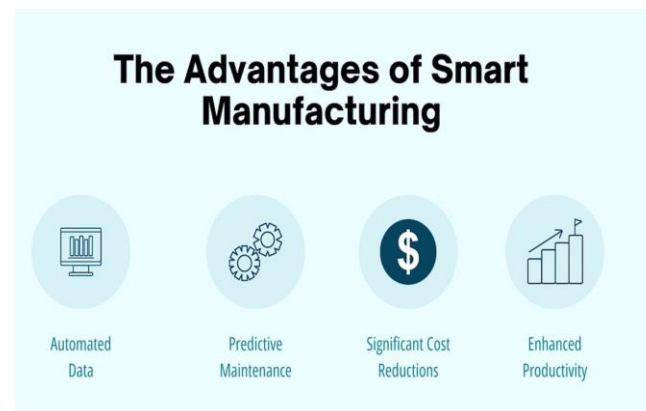
Transactional and Workflow Based Application

Typical Use Cases:

- SMART Home



- SMART Factory



Source: <https://www.businesswire.com/news/home/20201126005672/en/Why-Does-the-World-Need-Smart-Manufacturing-Now-Infinitis-Research-Experts-Discuss-and-Explain>

05. IIoT Use Cases

- Transactional and Workflow Based Application
- **Monitoring and Remote Control**
- Advanced Analytics/AI Application
- Sustainability Use Cases

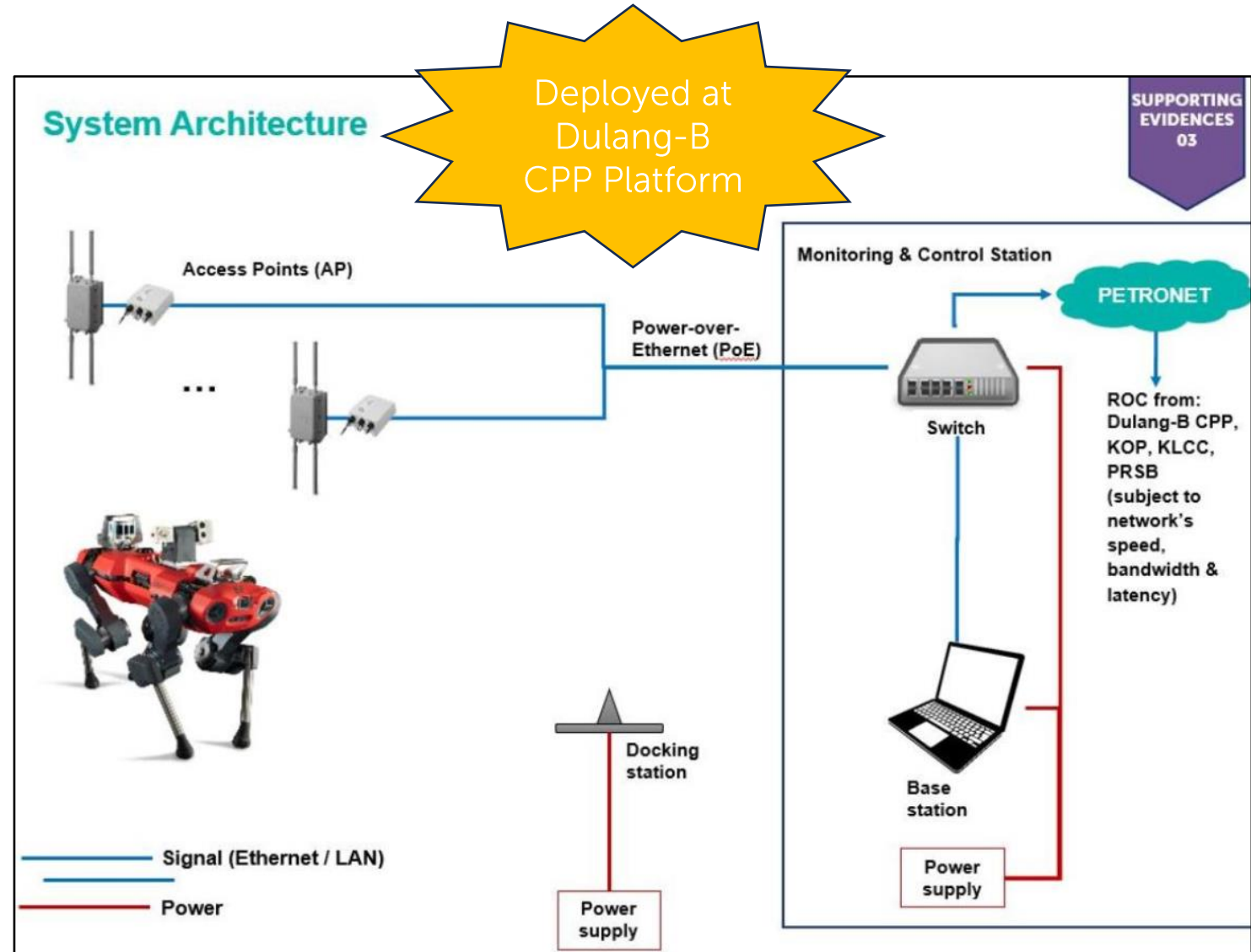
Monitoring and Remote Control

ANYmal ROBOT

ANYmal → robot deployed at site to perform surveillance and inspection

Advantages:

- 1) Safety Improvement
- 2) Continuous Autonomous Monitoring
- 3) Cost Efficiency
- 4) Integration with Digital Twin & AI



5.2 Monitoring and Remote Control

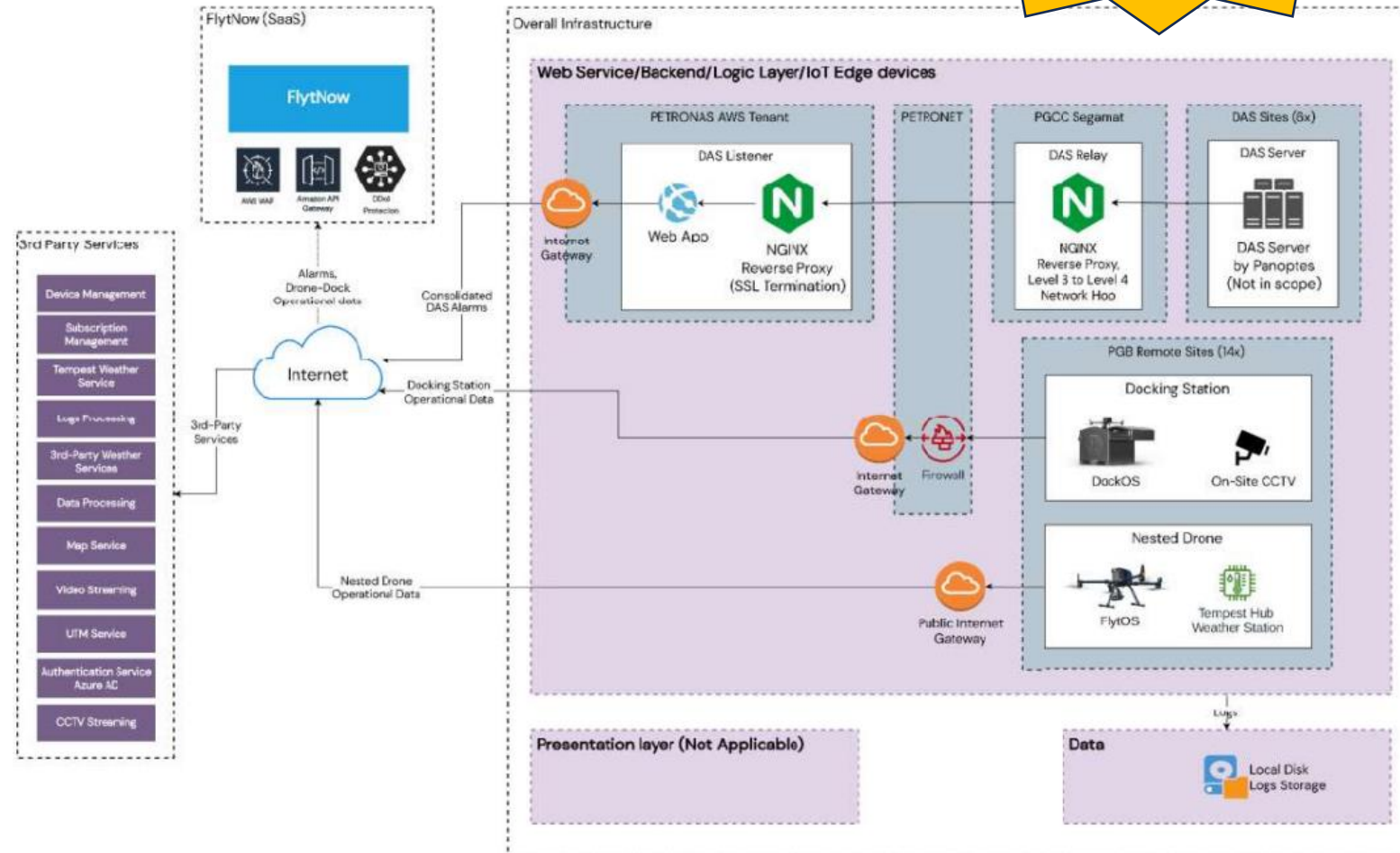
PGB Nested DRONE

Nested Drone → several drones housed in an autonomous docking station to provide continuous & automated monitoring

Deployed at
PGB

Advantages:

- 1) Continuous & Automated Monitoring
- 2) Enhanced Safety & Reduced Human Exposure
- 3) Cost Savings
- 4) High Resolution Data Collection
- 5) Adaptability to Harsh Environment
- 6) Quick Deployment & Scalability
- 7) Integration with Digital Twin & Remote Monitoring

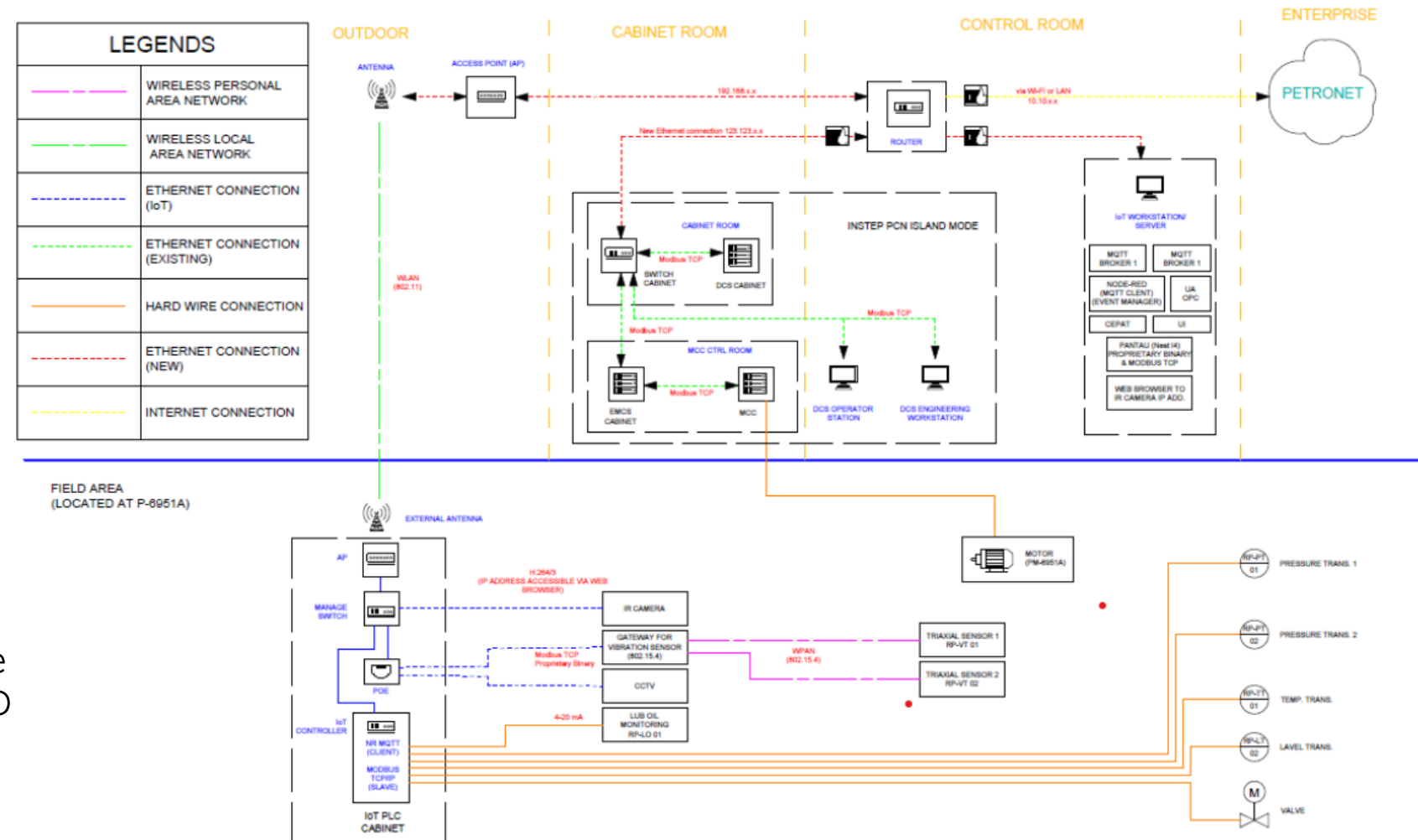


05. IIoT Use Cases

- Transactional and Workflow Based Application
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- **Advanced Analytics/AI Application**
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Advanced Analytics/AI Application

Rotating Equipment Package Remote Operation



Advantages:

- 1) Able to locate workers especially during Emergency/Turn around
- 2) Able to detect individual with close proximity with PUI ie during COVID time

Advanced Analytics/AI Application

Secure Personnel Online Tracking (SPOT)

System for contact tracing which consists of mobile application, tag device and data analytics



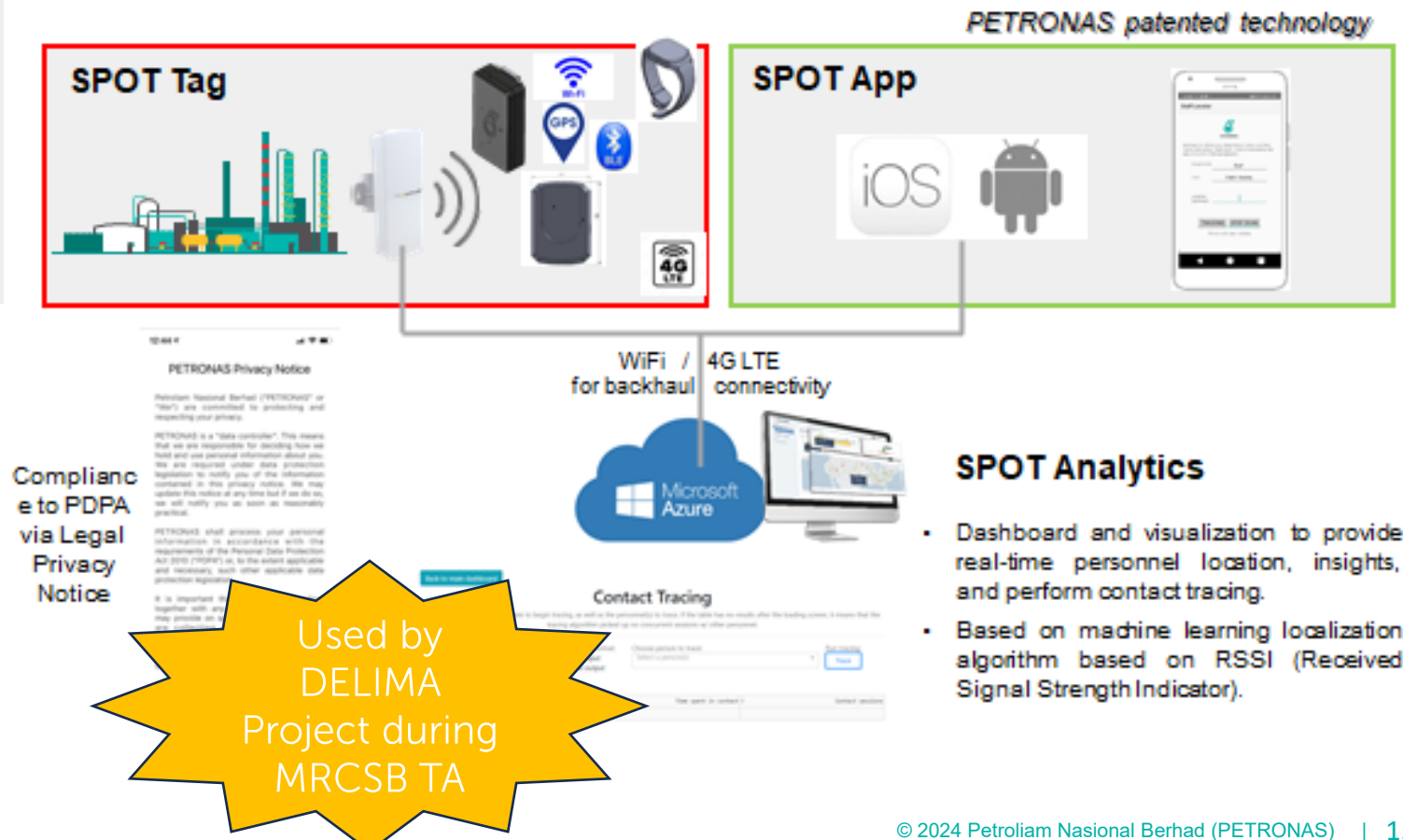
SPOT

Secure Personnel Online Tracking

Advantages:

- 1) Able to locate workers especially during Emergency/Turn around
- 2) Able to detect individual with close proximity with PUI ie during COVID time

SPOT technology comprises of mobile application (Android ready and incoming for iOS), tag device (non-Ex ready and incoming Q321 Ex certified) and data analytics software that provides AI & reports function.

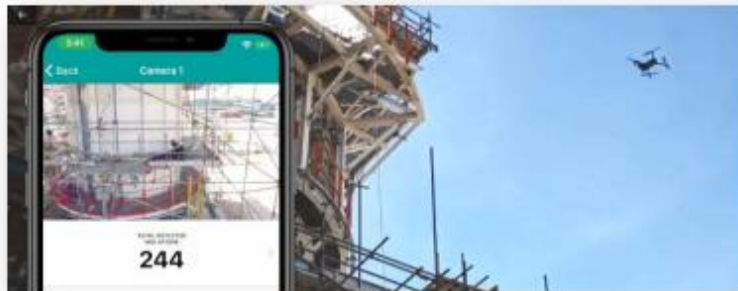


Advanced Analytics/AI Application

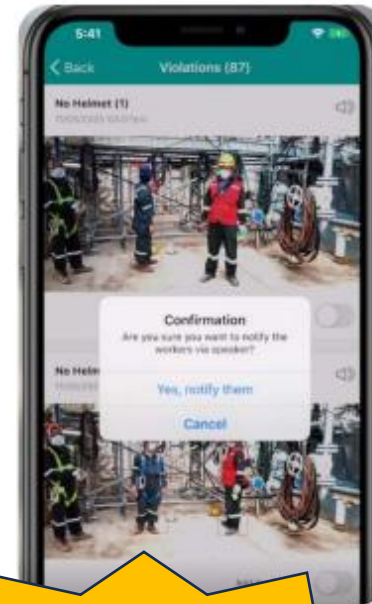
Artificial Intelligence Safety Camera (AISC)

AISC (Artificial Intelligence Safety Camera) is a technology developed for safety monitoring such as unsafe act and unsafe condition, Personal Protective Equipment ("PPE") detection as well as COVID-19 social distancing / mask requirements.

Video feeds from CCTVs, mobile Ex-certified cameras or drones are fed through the AI API on the server, which then flags non compliances and alerts HSSE supervisors through the app – greatly increasing their coverage



AISC technology presents numerous advantages over the current approach in which the monitoring is highly dependent on safety officers through actual monitoring at worksite or through live monitoring via Closed-circuit television ("CCTV").



Used by
DELIMA project
& MTBE



Social distancing can also be visually verified

Advanced Analytics/AI Application

Artificial Intelligence Safety Camera (AISC)

AISC : How It Works



- 1** Workers will be monitored continuously through PETRONAS AISC
- 2** Any HSE non-compliance will be detected and sent to AI server for analysis.
- 3** AI system will send onsite non-compliance evidence via push notification to mobile and desktop apps.
- 4** Safety officer immediately execute Proactive Incident Prevention Actions

AISC : Value Creation



Enable Proactive Incident Prevention through prompt response on HSE non-compliance



Magnify safety officer's capabilities



Digitize and Automate safety monitoring assurance

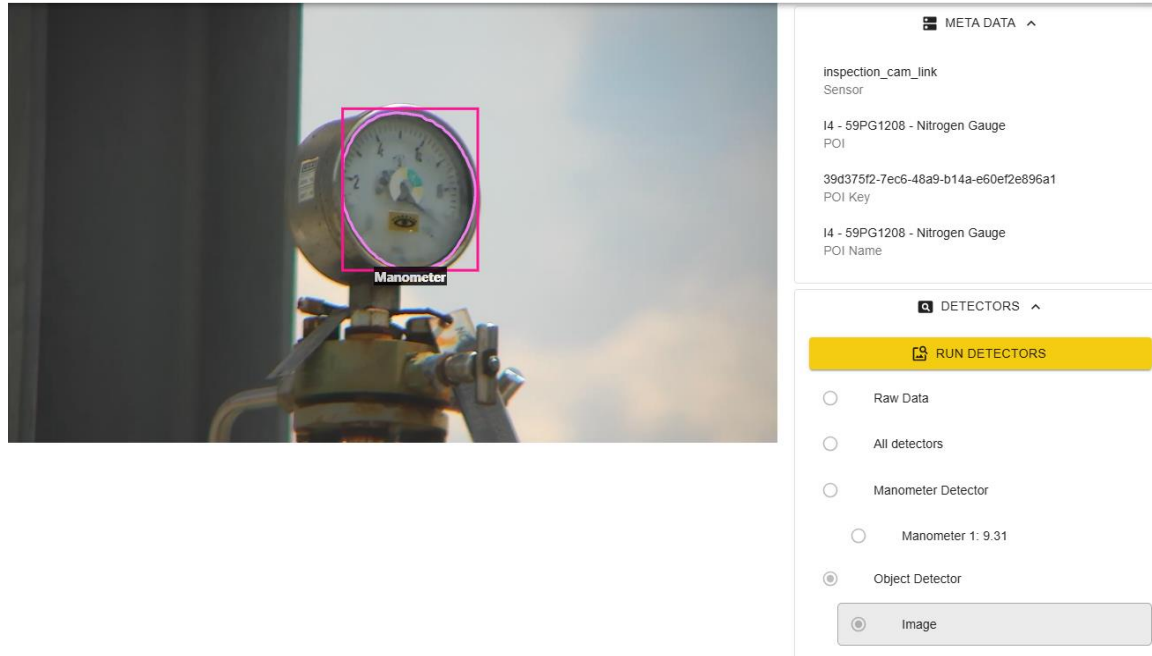


Increase conformance to HSE guidelines by 100% - upholding ZeTo Rules

Source: PDSB AISC

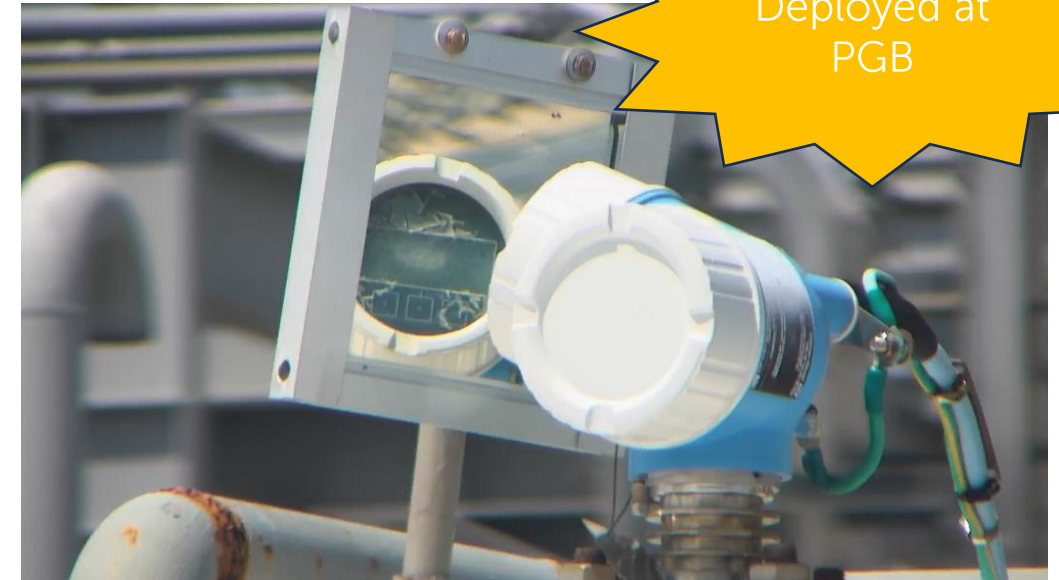
Advanced Analytics/AI Application

Computer Vision for Analogue Gauge



Advantages:

- 1) Local gauge can be monitored directly from control room
- 2) Devices with difficult remote/congested area can be monitored remotely/easily



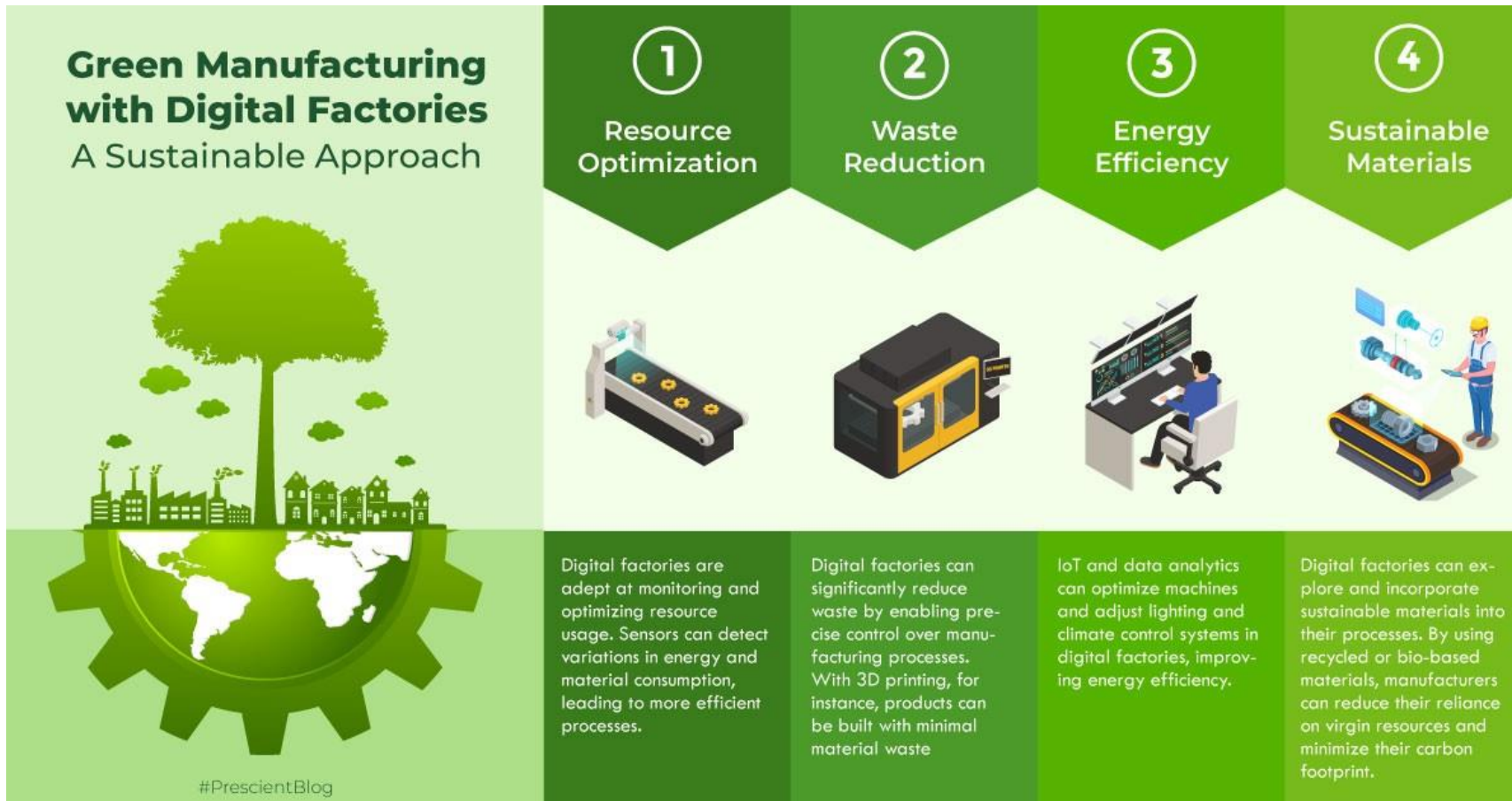
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Sustainability Use Cases

Green Manufacturing with Digital Factories

IIoT is one of the tools we can do wonderful things with. Its technology can be a game changer for economic, environmental and social sustainability in the industry.



Source: <https://www.linkedin.com/pulse/green-manufacturing-digital-factories-sustainable-yew7f/>

The IoT is an enabler for achieving the SDGs within the commercial framework. In phase one of this study (visit <http://wef.ch/IoT4D>) a database of over 640 IoT projects were mapped against 17 SDGs with the following results:

- 84% of the 643 analysed IoT deployments support or have the potential to support the SDGs
- 75% of these focus on five SDGs which are closer to commercial and industrial applications; SDG 9 (industry, innovation and infrastructure), SDG 11 (sustainable cities and communities), SDG 7 (affordable and clean energy); SDG 3 (good health and well-being), SDG 12 (responsible production and consumption)
- 95% of projects are small/medium-sized

The study supports the intuition many private sector executives share – that IoT has development benefits that could be maximized without compromising the commercial viability. The reason the IoT could become a game-changer for sustainability lies in its technology. At its core, IoT is about measuring and remotely controlling previously unconnected “things”. The technology reaches people and objects that technology could previously not reach.

Source:

<https://www3.weforum.org/docs/IoTGuidelinesforSustainability.pdf>

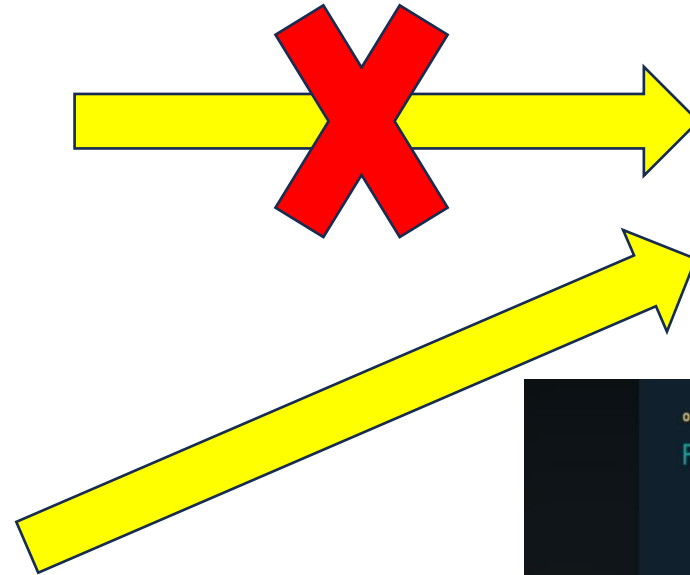
Sustainability Use Cases

ANYMal Testing at Dulang B

Example 1



- 1) Reduce diesel consumption
- 2) Reduce emission by boat



OFFSHORE FIELD TEST //

Field Test with ANYmal at Dulang B

Goal of the Field Test

- Robot locomotion at offshore environment & platform
- Autonomous navigation and inspection

Tasks for ANYmal

- Autonomous navigation:
 - Platform walk-about
 - Stair climbing
- Autonomous Inspections:
 - General Visual Inspection (GVI)
 - Thermal imaging
 - Audio recording
- Autonomous battery recharging

5 // ANYbotics // Dez-20

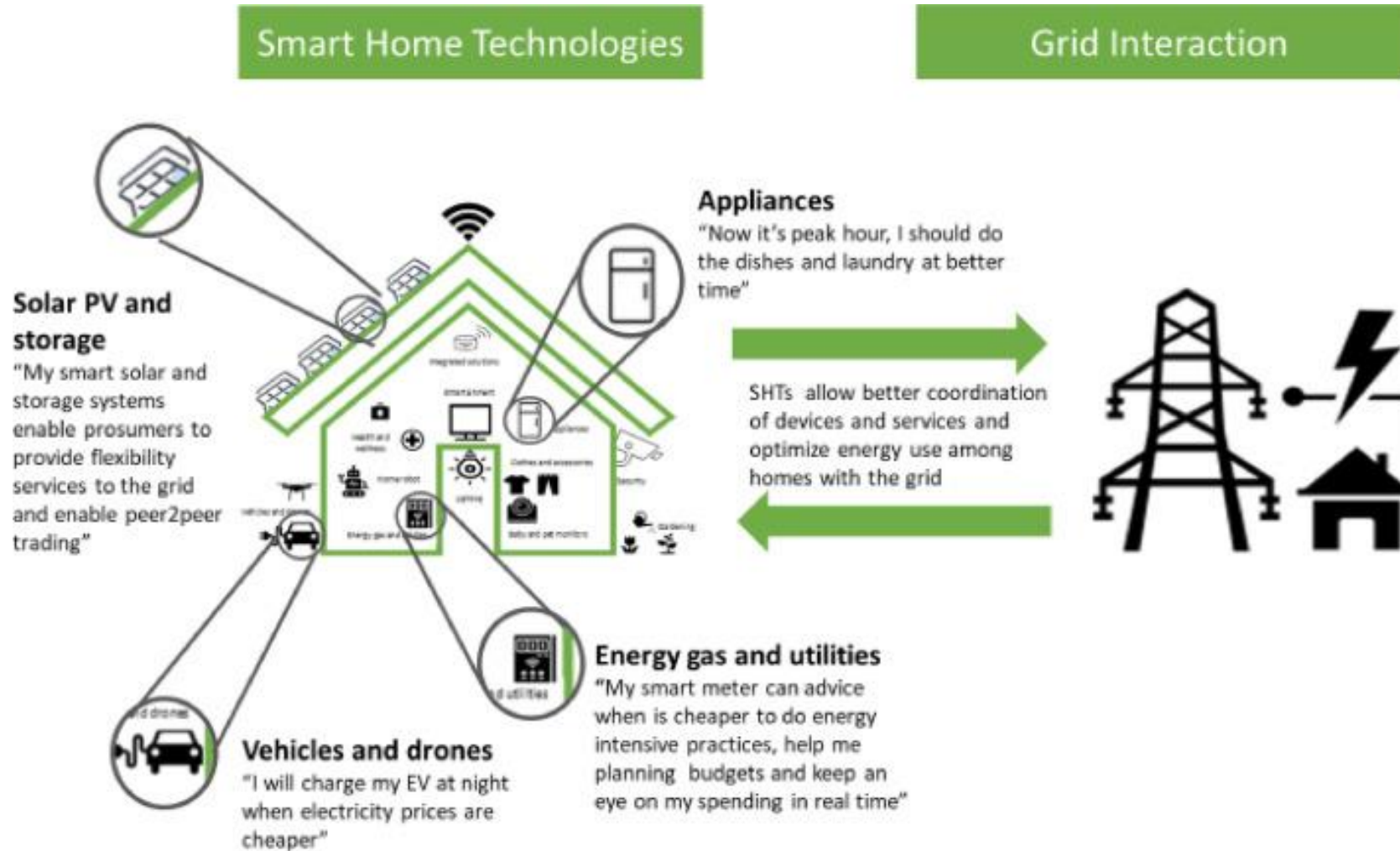
PETRONAS

ANYbotics

Sustainability Use Cases

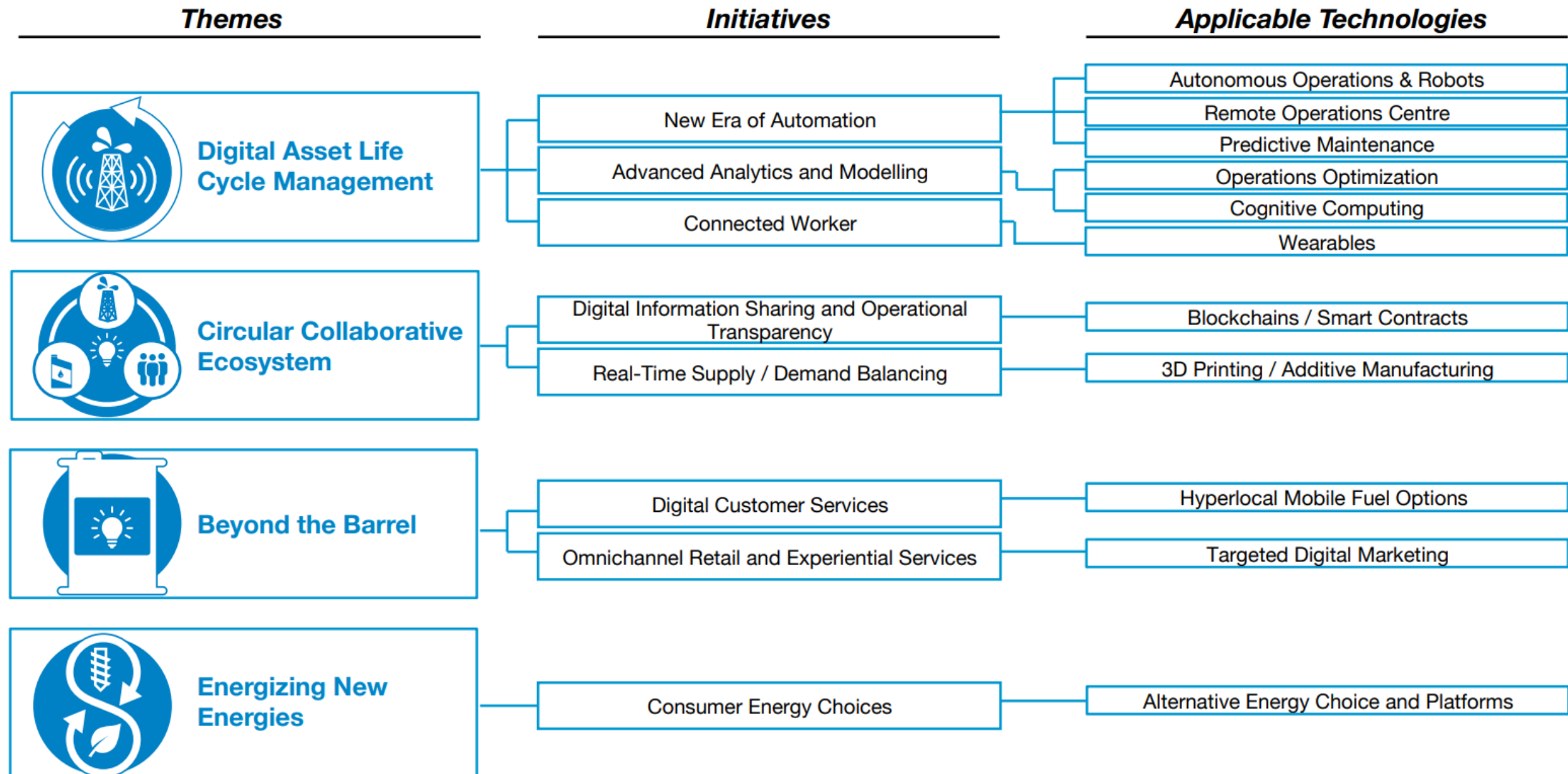
Smart Home Technologies

Example 2



Source: <https://www.sciencedirect.com/science/article/pii/S266627872100012X>

Sustainability Use Cases



Sustainability Use Cases

Sustainability Challenges: Digital Technology Adoption



- ✓ **Top challenge is implementation cost** in light of reduced margins and other cost pressures (e.g. infrastructure upgrades, employee training, external services)—challenges felt acutely by smaller businesses.
- ✓ **2nd most-cited challenge: digital skills:** There is a need to upskill the workforce due to scarcity of experts and project-delivery personnel (e.g. experienced in agile delivery methods) to implement advanced digital platforms.
- ✓ **3rd most common challenge is infrastructure:** Outdated platforms and server/network/portal capacity present constraints on advanced technology systems.

These challenges can be overcome through greater collaboration across the broader finance ecosystem enabled by digital technologies.

